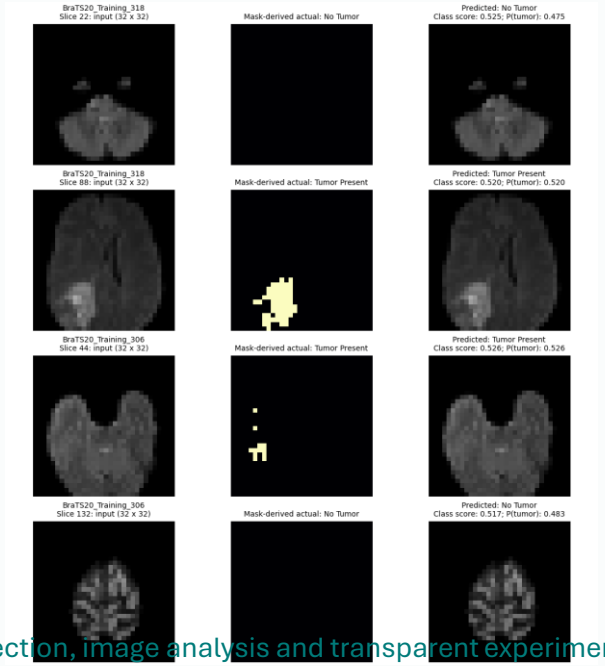
BRAIN + MRI RESEARCH PLATFORM

Brain Tumor Detection Using MRI Images

Brain + MRI Research Platform | Academic project presentation

- Evidence-led biomedical imaging, tutorial delivery and experimental AI workflow

- Research-only: no clinical diagnosis or patient-level claim



Objective: reproducible MRI inspection, image analysis and transparent experimental slice-level CNN inference.

Actual project evidence · source images, notebook output or saved model artifact

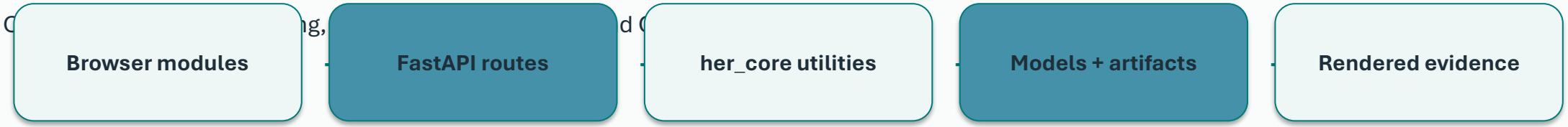
Actual CNN preview artifact.

BRAIN + MRI RESEARCH PLATFORM

Platform architecture

Independent browser modules reuse validated MRI and analytics utilities.

- Browser: Dashboard, Tutorial Lab, Image Analysis Studio, MRI Research Studio.
- FastAPI: validation, data-URL image serialization, case discovery and reporting.
- C… ng, …d C…



Data flow: validated local FLAIR/SEG → shared preprocessing → algorithm/checkpoint → compact PNG/metrics response → research UI.

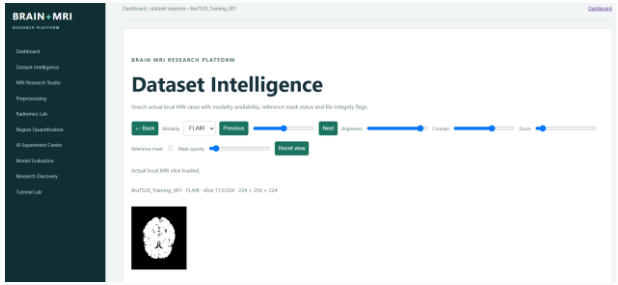
Research-only · not a clinical diagnosis

BRAIN + MRI RESEARCH PLATFORM

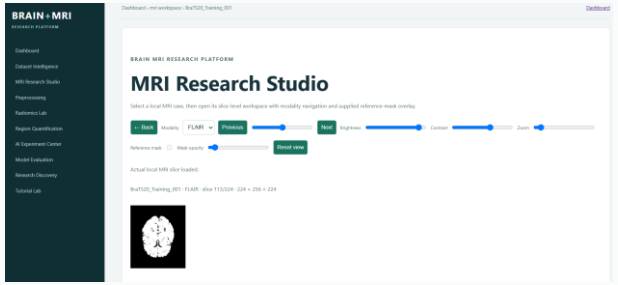
Frontend modules: contextual research workspace

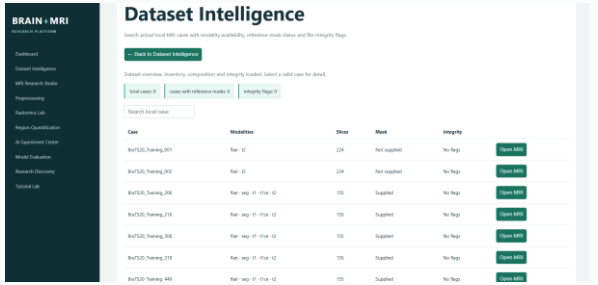
Actual application screens show Dataset Intelligence and MRI Research Studio.

- Dataset Intelligence provides local case selection, modality context and actual slice preview.
- MRI Research Studio separates workflow selection from case-specific analysis and upload inference.
- The visible Back action follows contextual module hierarchy rather than always returning to Dashboard.



Dataset Intelligence: inventory, case details and true local MRI availability.





Actual project evidence · source images, notebook output or saved model artifact

Actual frontend screenshots: Dataset Intelligence and MRI Research Studio.

BRAIN + MRI RESEARCH PLATFORM

Data lineage and safeguards

Every visible result has a traceable source and status.

- Local FLAIR/SEG → validate → preprocess → algorithm/checkpoint → PNG/metrics → UI.
- Missing masks stay missing; they are never converted to negative labels.
- The persistent checkpoint is reused; request handling never trains a new model.



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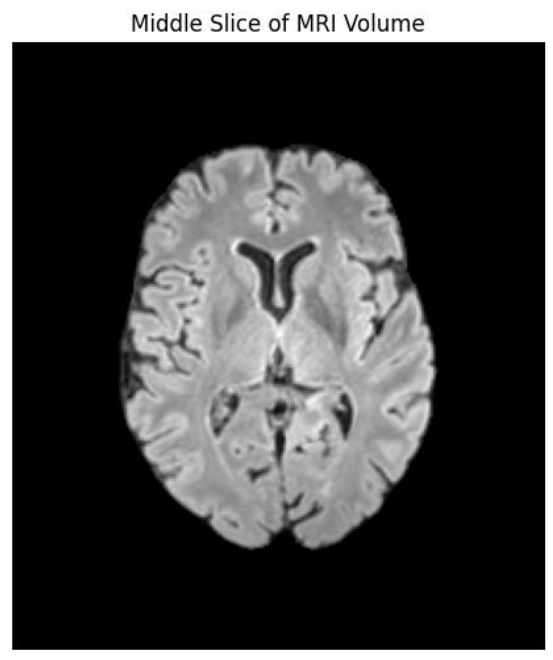
Dataset audit: local BraTS2020 evidence

Actual findings from CNN_DATASET_AUDIT.md.

- 8 case folders; FLAIR 8/8; SEG 6/8. Two unsegmented cases are excluded from supervision.

- Six FLAIR/SEG pairs are 240 × 240 × 155. Labels 0,1,2,3; y=1 iff mask > 0.

- All eligible labelled slices: 798 (355 negative, 443 positive



Original notebook output: local MRI middle slice.

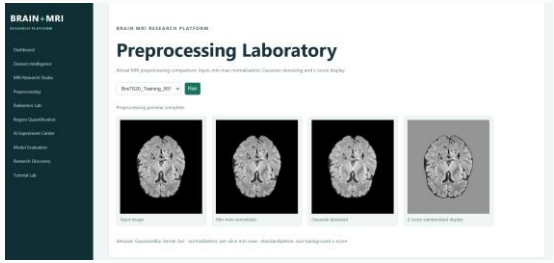
Actual project evidence · source images, notebook output or saved model artifact

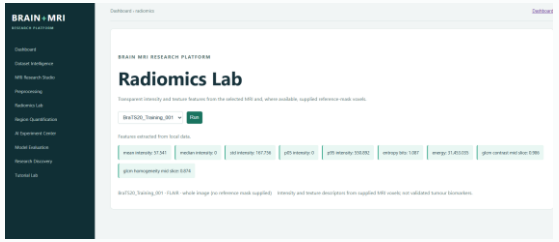
BRAIN + MRI RESEARCH PLATFORM

Implemented application analysis modules

Frontend outputs demonstrate that preprocessing and radiomics are separate functional modules.

- Preprocessing Laboratory renders original, cleaned, normalized and contrast-enhanced MRI stages.
- Radiomics Lab selects a case/modality and exposes feature-analysis actions backed by actual local data.
- These modules are analysis tools; they do not report clinical diagnoses.





Actual frontend screenshots: Preprocessing Laboratory and Radiomics Lab.

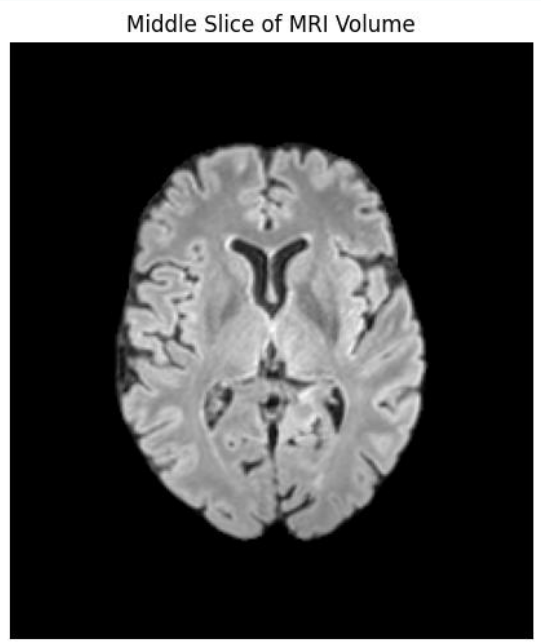
Actual project evidence · source images, notebook output or saved model artifact

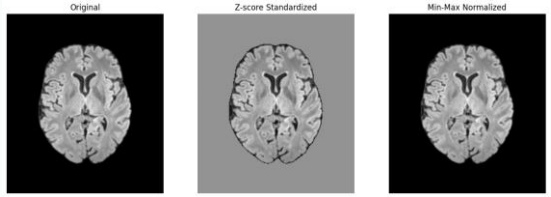
BRAIN + MRI RESEARCH PLATFORM

T1–T3: inspection, preprocessing and data quality

Foundational workflows establish data conditions before analysis.

- T1: discover local cases/modalities and show a middle slice.
- T2: clean invalid values; compare z-score and min–max transformations.
- T3: report missing modalities/duplicates without destructive assumptions.





Actual project evidence · source images, notebook output or saved model artifact

Original notebook evidence for T1/T2.

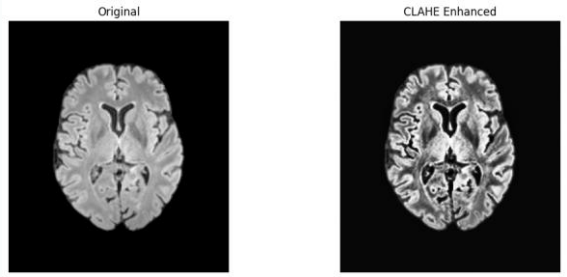
Research-only · not a clinical diagnosis

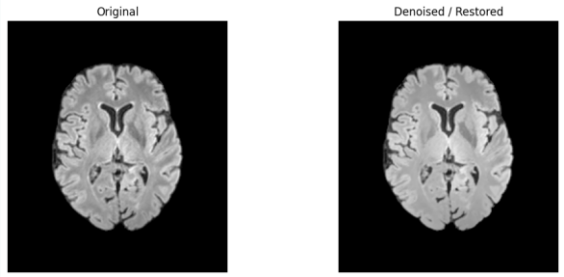
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T4: biomedical image analysis

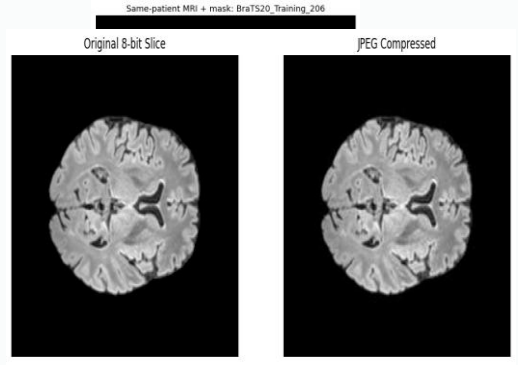
Aim: make enhancement and quality operations inspectable.

- Input: local MRI slice. CLAHE enhancement and bilateral denoising are displayed side by side.
- Interpretation: altered contrast is presentation support—not tumour segmentation/diagnosis.
- Implementation: notebook operations and reusable preprocessing helpers.





Reference-mask overlay



JPEG compression comparison

Actual T4 notebook outputs: enhancement and restoration.

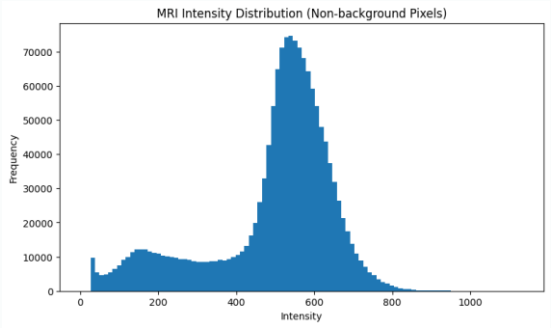
Actual project evidence · source images, notebook output or saved model artifact

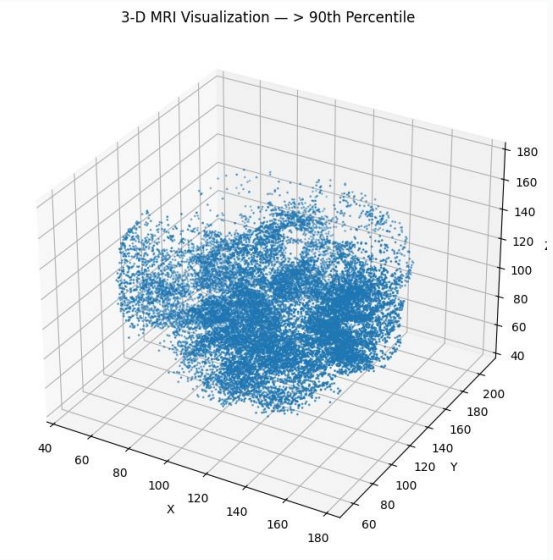
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T4: intensity and 3-D context

Additional notebook evidence links slice-level work to volume context.

- Histogram uses non-background MRI intensity values, making intensity distribution explicit.
- 3-D threshold display is a visualisation of high-intensity voxels, not a tumour segmentation.
- These figures support inspection before feature extraction or classification.





Actual project evidence · source images, notebook output or saved model artifact

Original notebook intensity and 3-D visualisation outputs.

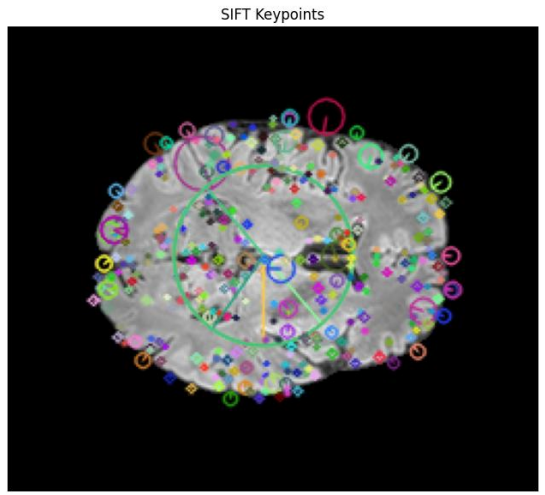
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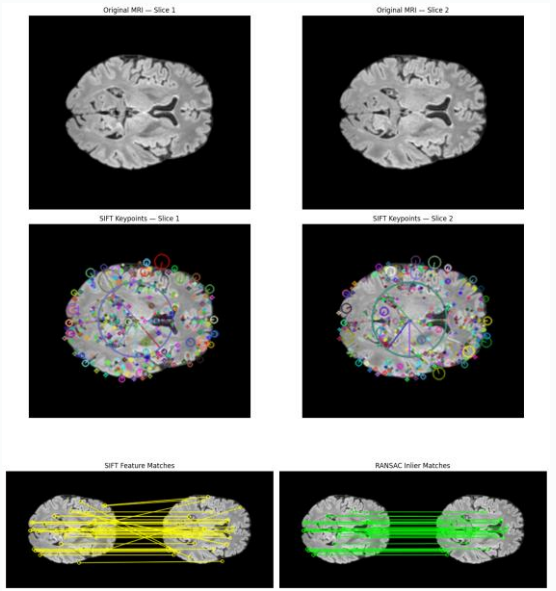
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T5: SIFT + RANSAC

Aim: robust correspondence evidence between nearby MRI slices.

- SIFT detects scale-invariant keypoints and descriptors from normalized intensity.
- Descriptor matching proposes candidates; RANSAC estimates a robust transform/rejects outliers.
- Output supports image correspondence analysis, not tumour presence.





Actual project evidence · source images, notebook output or saved model artifact

Actual T5 keypoint and RANSAC-match outputs.

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BRAIN + MRI RESEARCH PLATFORM

T5 CNN: target and leakage control

Binary slice labels derive only from reference-mask pixels.

- y = 1 if any(mask > 0), otherwise 0; unlabelled cases are excluded.
- Case separation: train 4 cases/132 slices; validation 1/35; held-out test 1/35.
- N par



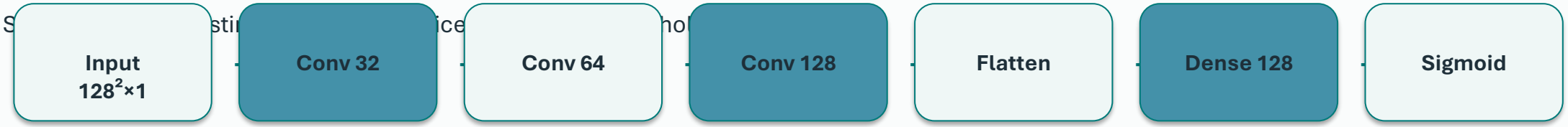
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BRAIN + MRI RESEARCH PLATFORM

T5 CNN: architecture and experiment configuration

Saved model brats-flair-slice-cnn-v1 | input 128 × 128 × 1.

- Conv2D(32,3×3,ReLU,same) + pool → Conv2D(64) + pool → Conv2D(128) + pool.
- Flatten → Dense(128, ReLU) → Dropout(0.5) → Dense(1, sigmoid).
- S sti ice ho



Input 128×128×1 FLAIR · Adam + binary cross-entropy · fixed seed 42 · 12 requested epochs · class weights 1.1786 / 0.8684

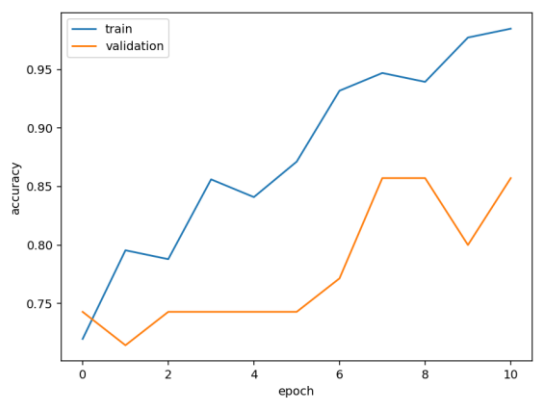
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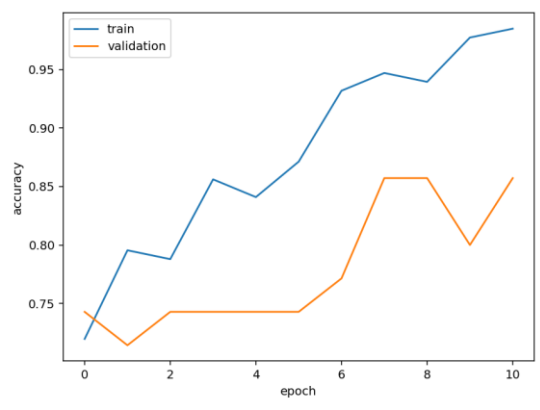
T5 CNN: training and validation accuracy

Actual persisted history; 11 recorded epochs.

- Training accuracy: 0.7197 → 0.9848. Validation ends at 0.8571.
- The gap is a small-cohort caution, not deployment evidence.
- Validation is for model selection; held-out reporting is separate.



Training accuracy



Training / validation loss

Actual project evidence · source images, notebook output or saved model artifact models/training_accuracy.png

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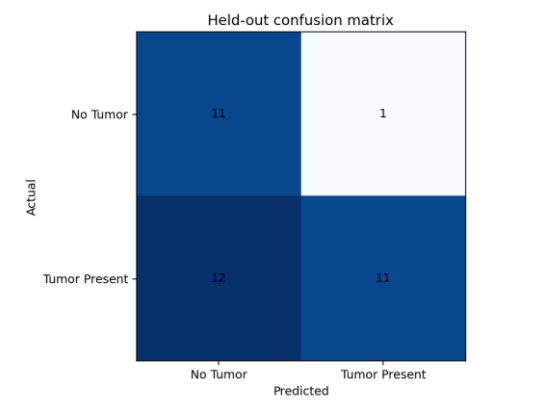
T5 CNN: held-out confusion matrix

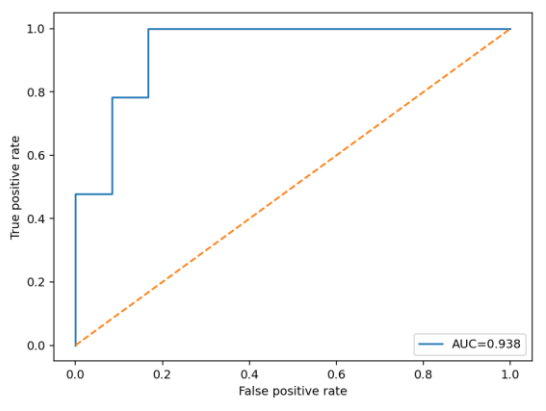
Actual one-patient held-out test: 35 slices.

- [[11,1],[12,11]] = 11 TN, 1 FP, 12 FN, 11 TP.

- Tumour recall = 0.4783: 12/23 positive slices missed at threshold 0.50.

- Negative slice result cannot rule out tumour.





Confusion matrix ROC curve (AUC = 0.9384)

Actual project evidence · source images, notebook output or saved model artifact models/confusion_matrix.png

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T5 CNN: classification report

Actual held-out metrics (35 slices).

Class	Precision	Recall	F1	Support
No Tumor	0.4783	0.9167	0.6286	12
Tumor Present	0.9167	0.4783	0.6286	23
Macro average	0.6975	0.6975	0.6286	35
Weighted average	0.7664	0.6286	0.6286	35

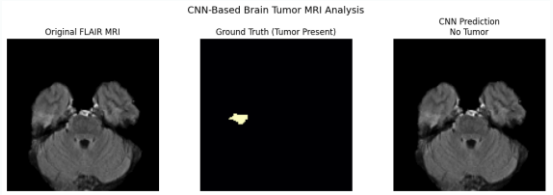
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BRAIN + MRI RESEARCH PLATFORM

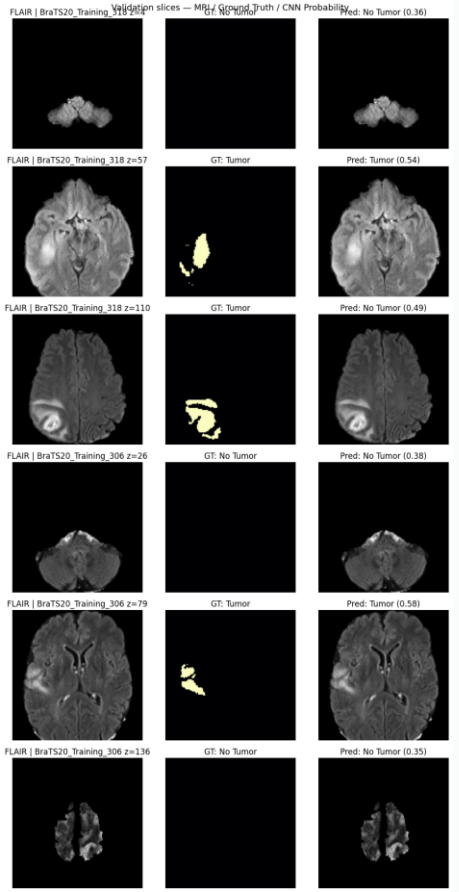
T5 CNN: actual inference visualization

Notebook evidence demonstrates genuine preprocessed-slice predictions.

- UI renders original MRI and exact 128 × 128 model input.
- Tutorial data can show actual label separately; arbitrary uploads do not have ground truth.
- Probabilities are model estimates, never diagnostic certainty.



Actual project evidence · source images, notebook output or saved model artifact



Original notebook CNN prediction outputs.

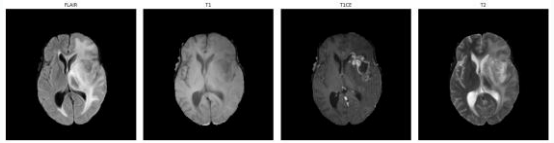
Research-only · not a clinical diagnosis

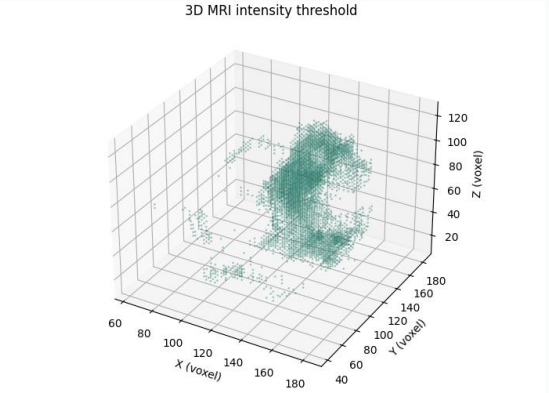
BRAIN + MRI RESEARCH PLATFORM

T6: multimodal MRI visualization

Aim: compare compatible modalities and supplied segmentation context.

- Same-case volumes are rendered with consistent orientation.
- SEG pane is supplied reference where present—not CNN output.
- Modality comparison precedes downstream interpretation.





Actual project evidence · source images, notebook output or saved model artifact

Actual T6 notebook/project visualizations.

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T7: MRI review queue

Aim: operational case review from actual data availability.

- Input: discovered local cases plus FLAIR/SEG availability.
- Output: review state and completeness status; T7–T9 do not request synthetic text examples.
- Availability is operationa



Actual project evidence · source images, notebook output or saved model artifact

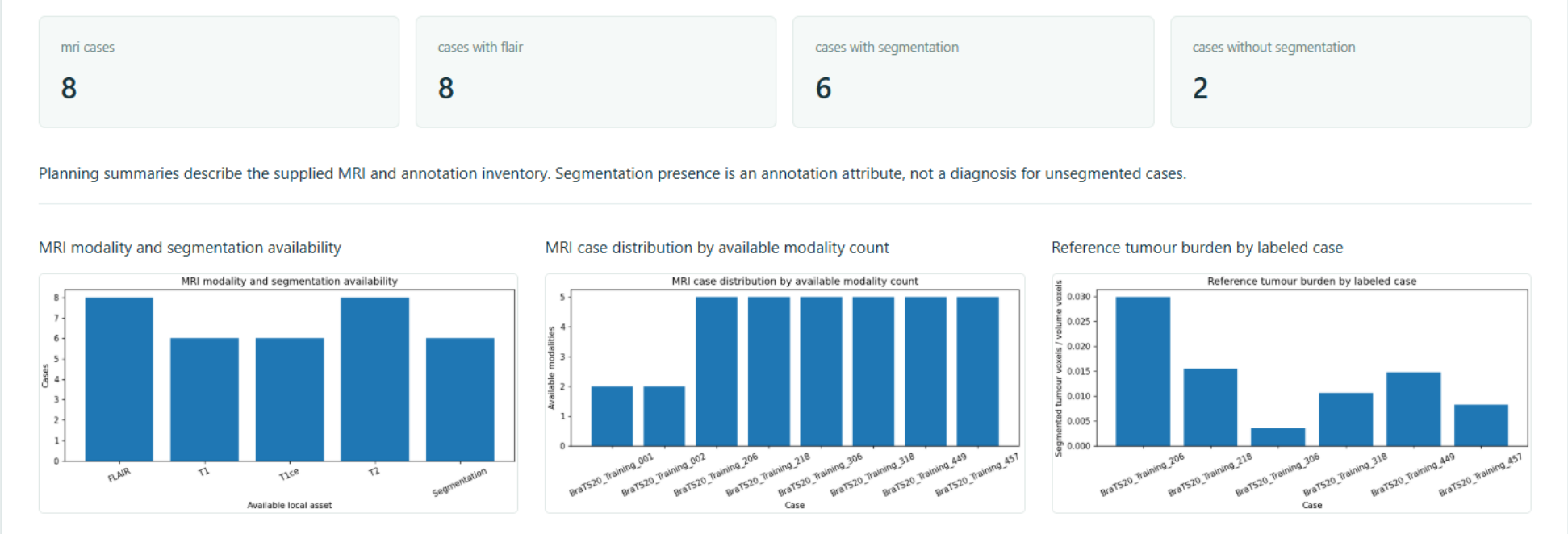
Actual T7 availability/completeness output.

BRAIN + MRI RESEARCH PLATFORM

T8: healthcare data analysis

Aim: inspect dataset inventory and supplied tumour-burden information.

- Inputs: local metadata and SEG only where provided.
- Processing: counts, availability checks and volume-derived summaries.
- Small loc



Actual T8 inventory and distribution dashboard.

Actual project evidence · source images, notebook output or saved model artifact

BRAIN + MRI RESEARCH PLATFORM

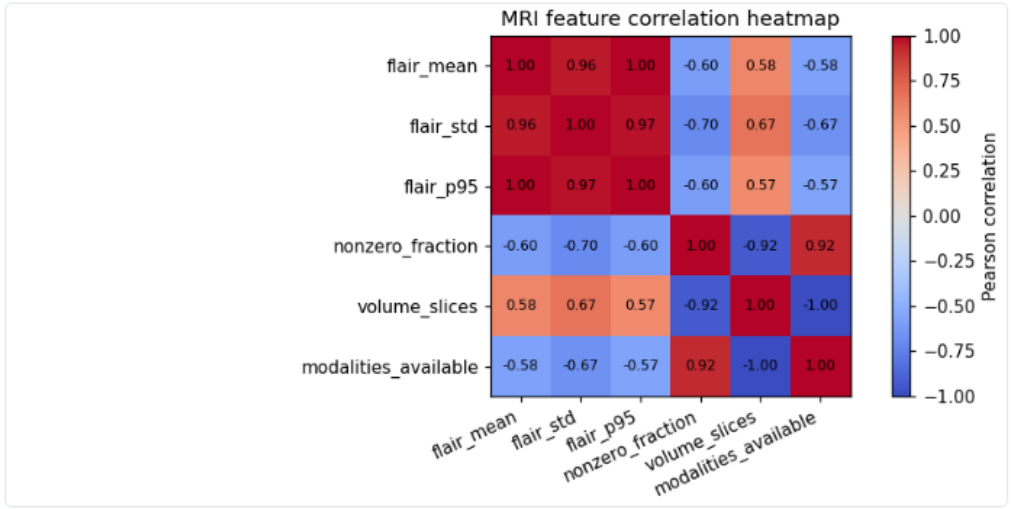
T9: feature exploration

Aim: derive MRI features for descriptive statistics.

- Normalized images/volumes yield intensity and simple morphology-derived exploration.

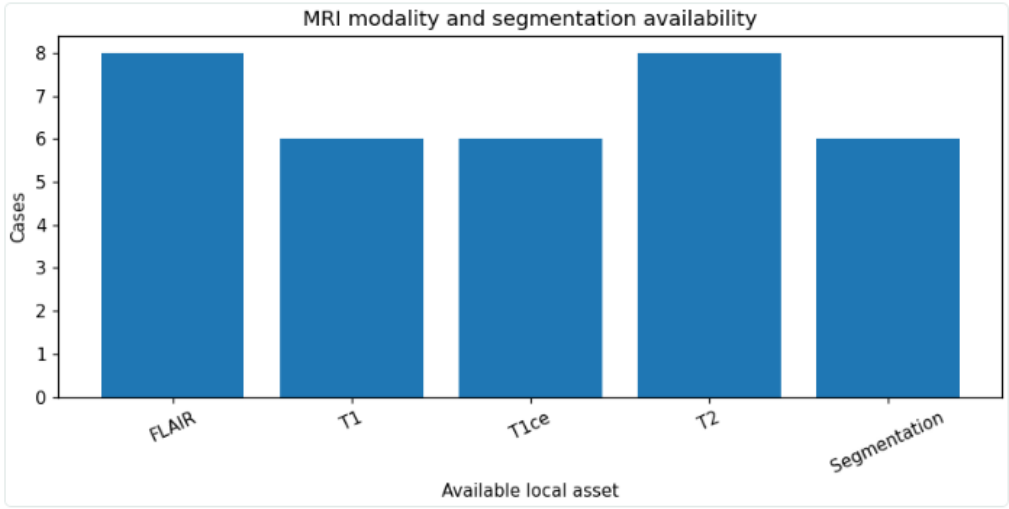
- Associations are descriptive; local cohort size limits inference and generalisation.

MRI feature correlation heatmap



Pairwise correlations of MRI-derived features. Small case count prevents confirmatory inference.

MRI modality and segmentation availability



Availability is read from local case folders; missing assets are not imputed.

Actual T9 feature-correlation and availability output.

Actual project evidence · source images, notebook output or saved model artifact

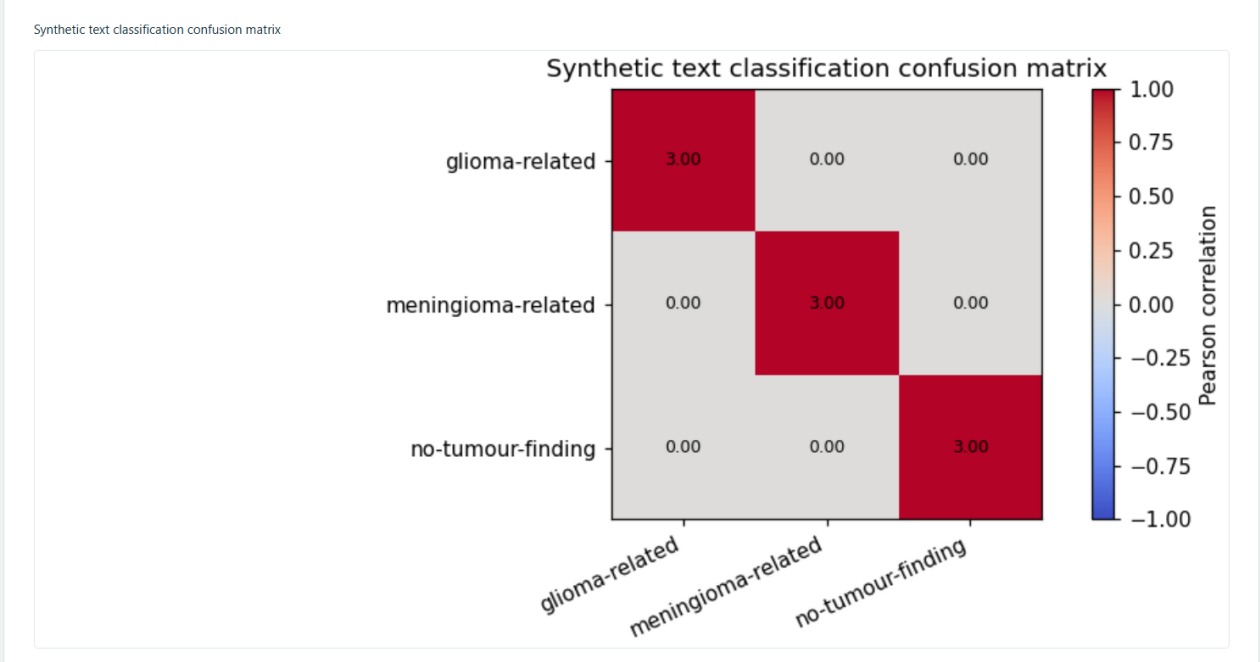
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T10: text classification

A synthetic teaching-report workflow, distinct from MRI imaging.

- Supported synthetic CSV examples are restricted to T10–T12.
- TF–IDF features feed logistic classification and class-oriented tutorial outputs.
- No real patient reports, clinical N



Actual T10 synthetic-text classification confusion matrix.

Actual project evidence · source images, notebook output or saved model artifact

Research-only · not a clinical diagnosis

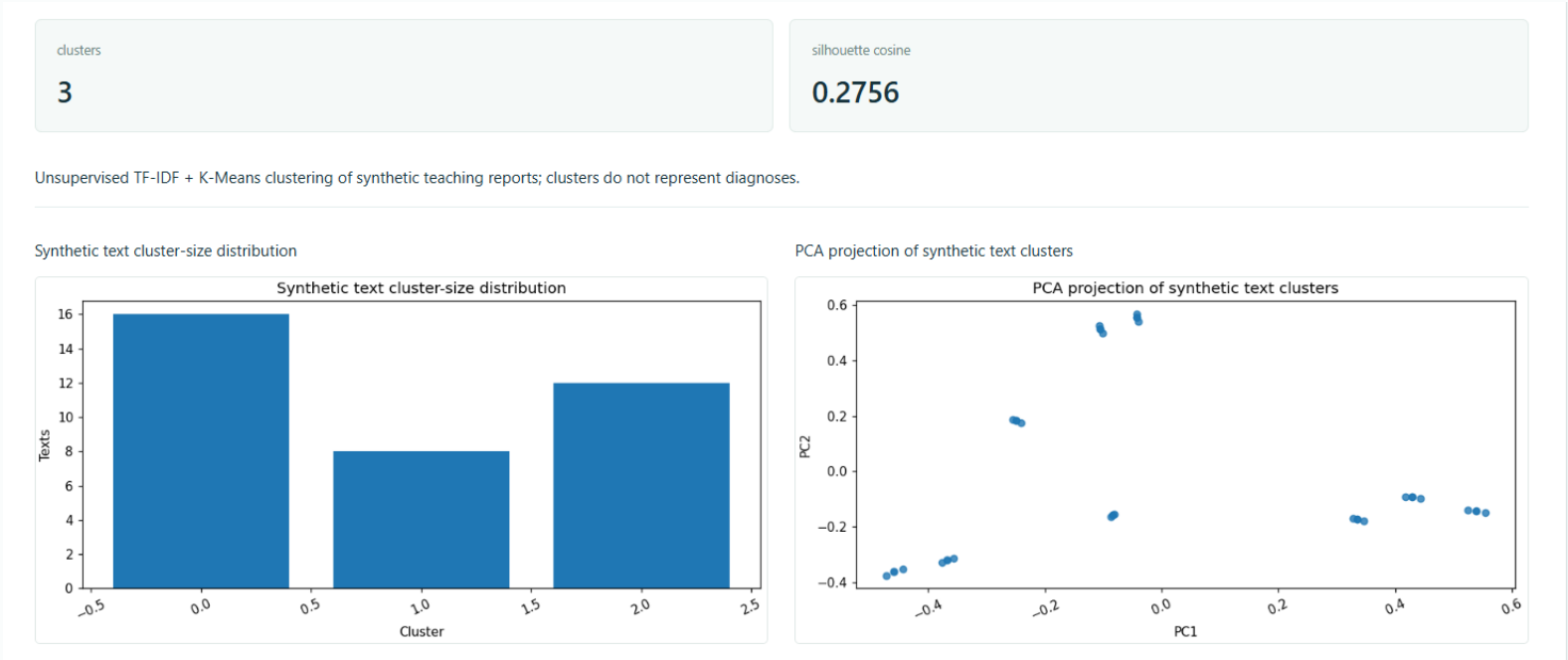
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T10: clustering

Clustering remains independent from text classification.

- TF–IDF vectors feed K-Means; cluster IDs describe lexical similarity, not diagnoses.

- Separate UI controls preserve the difference between supervised and unsupervised workflows.



Actual project evidence · source images, notebook output or saved model artifact

Actual T10 clustering distribution and PCA projection.

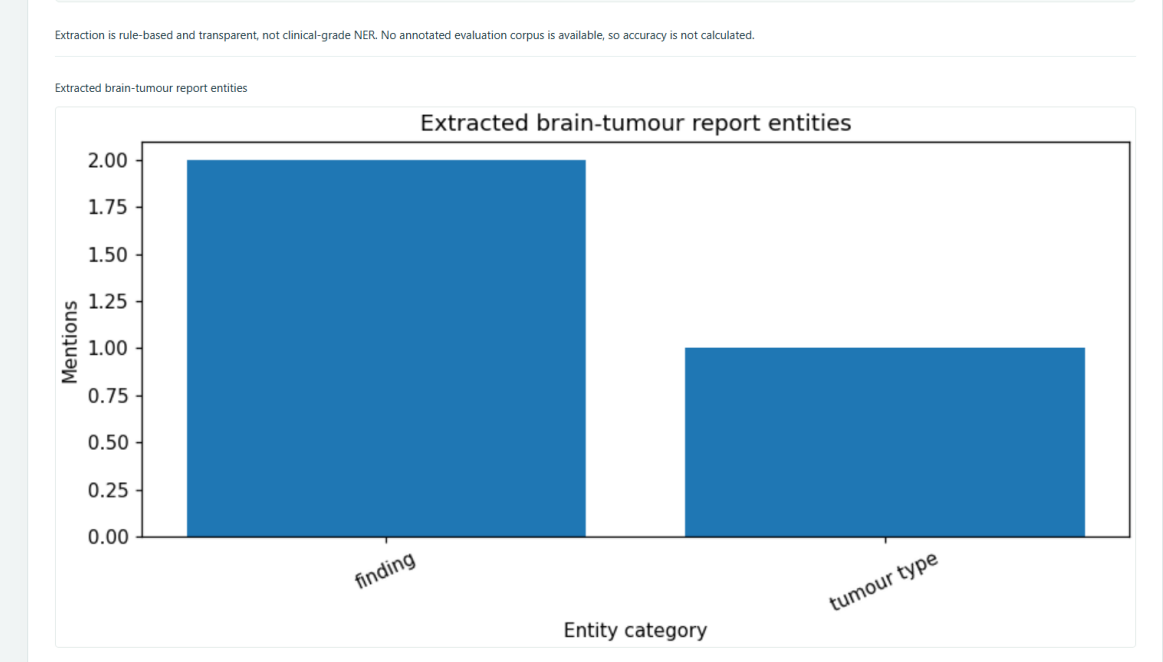
BRAIN + MRI RESEARCH PLATFORM

T11: entity extraction

Transparent lexicon and measurement-rule extraction.

- Synthetic text is processed with term and pattern matching.

- Output is structured teaching evidence, not a trained clinical NER system.



Actual T11 extracted-entity output.

Actual project evidence · source images, notebook output or saved model artifact

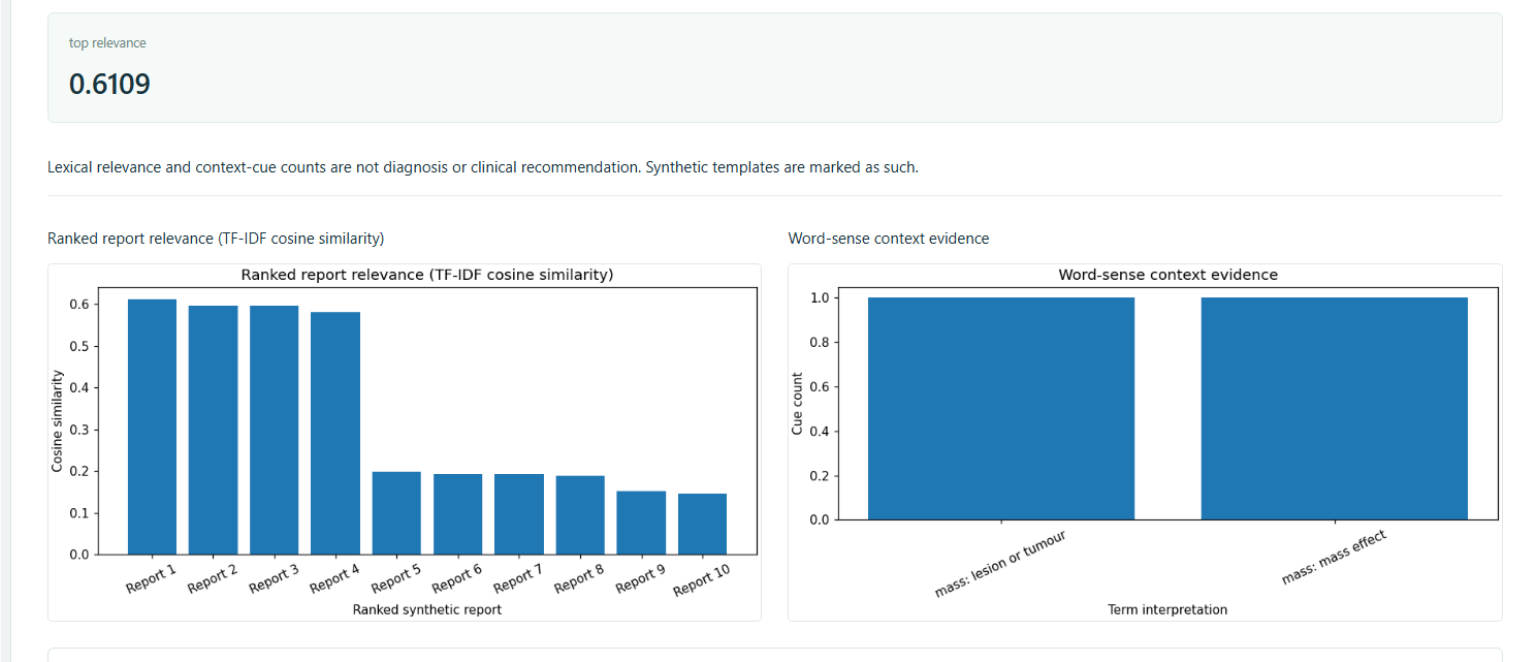
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T12: ranking and disambiguation

TF–IDF cosine ranking of synthetic report passages.

- Query/corpus → vectorisation → cosine similarity → ranked result with ambiguity context.

- This demonstrates retrieval mechanics, not clinical decision support.



Actual project evidence · source images, notebook output or saved model artifact

Actual T12 ranking and word-overlap output.

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T13: PCA and t-SNE

Aim: lower-dimensional exploratory projections of standardised MRI features.

- PCA reports linear projection/explained-variance evidence; t-SNE offers nonlinear neighbourhoods.

- Visual clusters are hypothesis-generating, not validated biological subtypes.



Actual project evidence · source images, notebook output or saved model artifact

Actual T13 PCA, t-SNE and feature-space output.

Research-only · not a clinical diagnosis

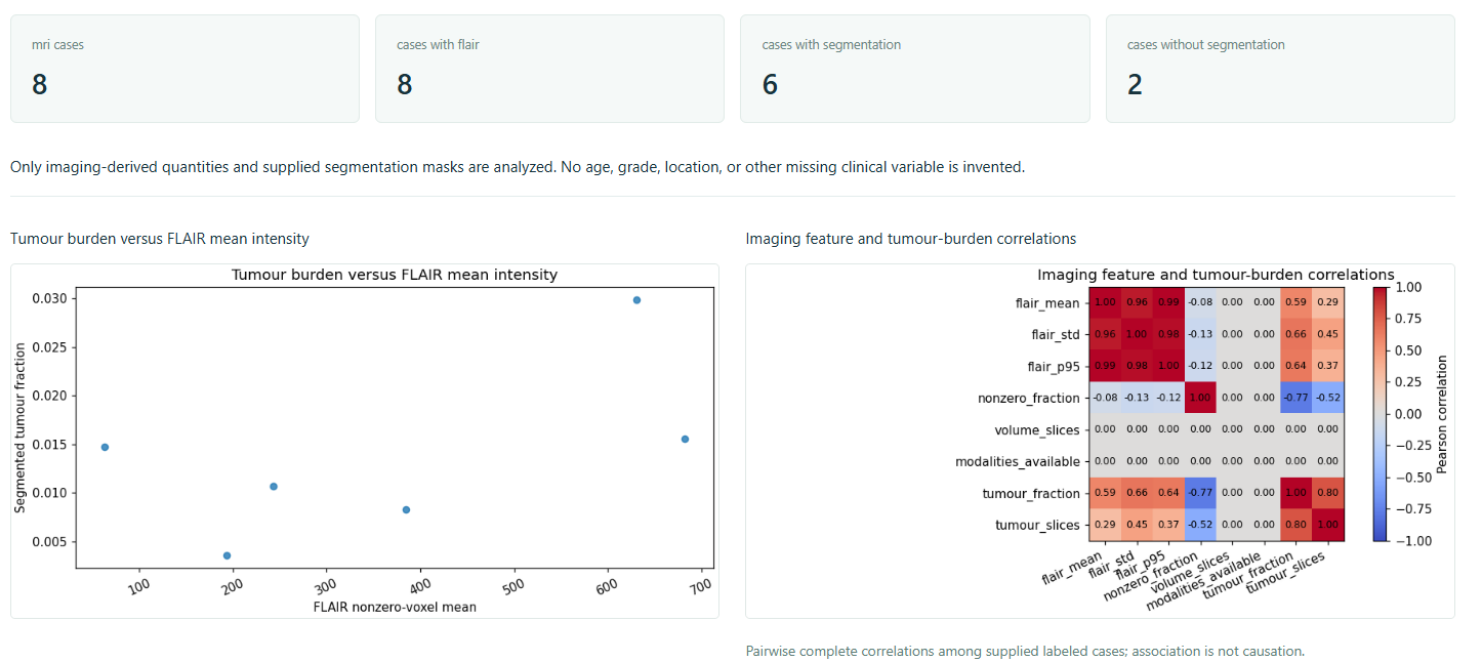
BRAIN + MRI RESEARCH PLATFORM

T14: tumour-burden associations

Aim: exploratory associations with supplied reference-mask burden.

- Feature summaries are paired with SEG-derived burden only when a reference mask exists.

- Association plots are non-causal and missing masks remain missing.



Actual project evidence · source images, notebook output or saved model artifact

Actual T14 association/correlation output.

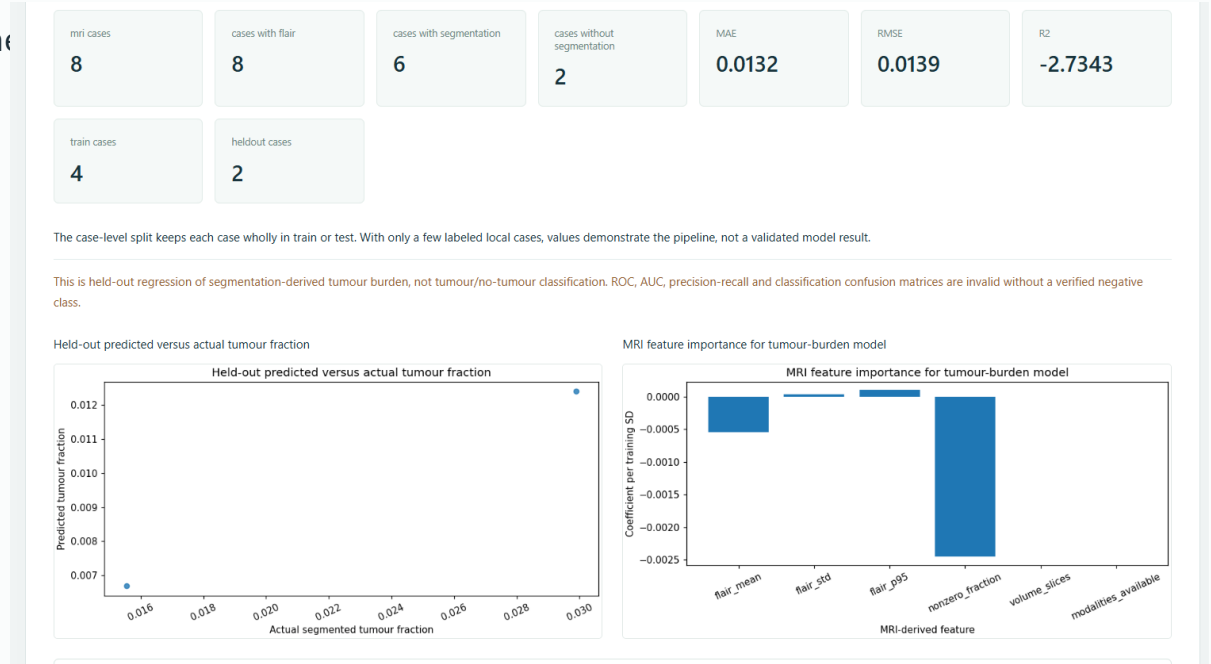
Research-only · not a clinical diagnosis

BRAIN + MRI RESEARCH PLATFORM

T15: held-out regression

Aim: estimate segmentation-derived tumour burden continuously.

- Task is regression, not tumour/no-tumour classification.
- Prediction-versus-reference visualisation requires regression metrics—not an invalid ROC claim.
- Local size limits confidence in gene



Actual project evidence · source images, notebook output or saved model artifact

Actual T15 held-out regression output.

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BRAIN + MRI RESEARCH PLATFORM

MRI Upload & AI Analysis

Persistent checkpoint inference is a separate application workflow.

- Compatible PNG/JPEG/NIfTI FLAIR inputs are server-validated (30 MB maximum).
- L/LA/RGB/RGBA images normalise safely to one grayscale channel, then reuse the unchanged FLAIR preprocessing.
- Response includes previews



Report outputs retain actual input metadata, previews, returned slice probabilities, threshold, duration and research-only limitations.

Actual MRI Upload & AI Analysis inference interface.

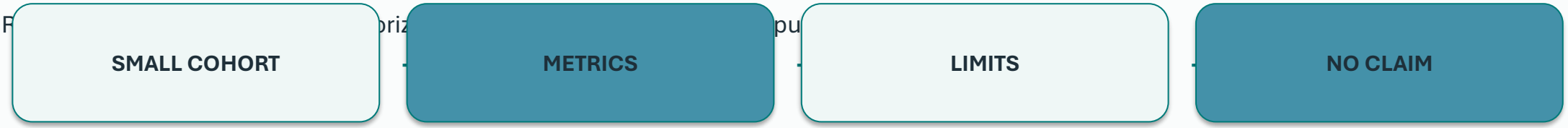
Actual project evidence · source images, notebook output or saved model artifact

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Limitations and safeguards

Transparency is a core technical requirement.

- Six labelled cases and one held-out test patient: experimental slice-level evidence only.
- No patient diagnosis, segmentation prediction, clinical confidence or arbitrary-upload accuracy is asserted.
- R… …ori… …pu…

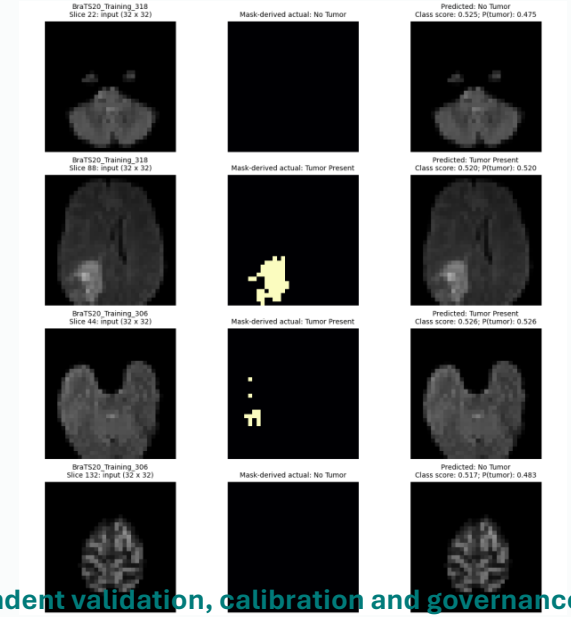


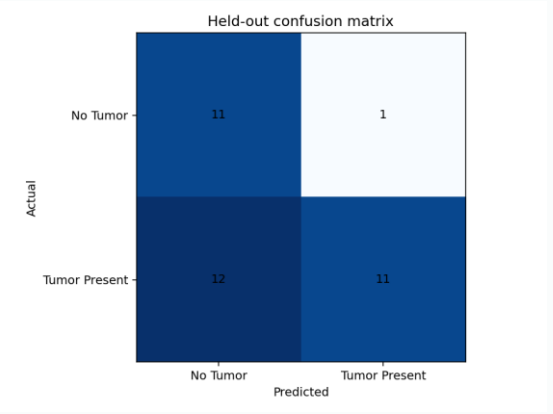
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Conclusion

A reproducible academic platform connecting MRI engineering, tutorials and transparent AI evidence.

- T1–T15 are represented through distinct aims, methods, actual output and limitations.
- The persistent FLAIR CNN is integrated but remains experimental and slice-level.
- Source-backed visuals and metrics make the work inspectable for an academic demonstration.





Future work: larger multi-centre cohorts, patient-level independent validation, calibration and governance before clinical translation.

Actual project evidence · source images, notebook output or saved model artifact

Actual inference and held-out evaluation artifacts.